26PDPA0 – Data Modelling and Visualization

Assignment 2a - MongoDB Exercises: (CO4)

Q1. Structure of 'restaurants' collection:


```
{
 "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
 },
 "borough": "Bronx",
 "cuisine": "Bakery",
 "grades": [
{ "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
{ "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
{ "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
{ "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
{ "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
 ],
 "name": "Morris Park Bake Shop",
 "restaurant_id": "30075445"
}
```


Install MongoDB and give the output of theses queries:

1. Write a MongoDB query to display the fields restaurant_id, name, borough and zip code, but exclude the field _id for all the documents in the collection restaurant.
2. Write a MongoDB query to display the first 5 restaurants which is in the borough Bronx.
3. Write a MongoDB query to find the restaurants that achieved a score, more than 80 but less than 100.
4. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which contain 'Reg' as three letters somewhere in its name.
5. Write a MongoDB query to arranged the name of the cuisine in ascending order and for that same cuisine borough should be in descending order.
6. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5.
7. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan.
8. Write a MongoDB query to find the highest score for each restaurant.
9. Write a MongoDB query to find the name and address of the restaurants that have at least one 'A' grade and one 'B' grade.
10. Write a MongoDB query to find the restaurant that has the highest average score for the cuisine "Turkish".

Q2. An online learning platform offers courses in programming, data science, cybersecurity, and cloud computing. Each course contains a title, description, instructor details, prerequisites, and a list of learning modules. Each module may include videos, reading materials, quizzes, and assignments. Since courses have different structures and learning resources, the platform requires a flexible data storage model. Design a sample document data model to represent a course and its associated learning modules and resources.

Q3. A railway reservation system processes a large number of bookings, cancellations, and ticket modifications every day. Each transaction includes a passenger ID, booking reference, train number, journey date, transaction timestamp, and transaction status. The system must retrieve a passenger's recent transactions and booking history without scanning the entire database. Design a wide-column data model to represent passenger transaction history, specifying appropriate keys and columns for efficient retrieval.

Q4. A food delivery application connects customers, restaurants, food items, and delivery partners. Customers place orders from restaurants, restaurants offer multiple food items, and delivery partners fulfil orders. Customers may also rate restaurants and food items. The application wants to identify restaurants associated with a customer's previous orders and suggest similar options. Design a graph data model by identifying the required node types, relationship types, and relevant properties.